Solar Energy

Chapter 7

Most energy on Earth begins with the sun. The sun transfers light and heat to the geosphere, atmosphere, hydrosphere, and biosphere. Unequal heating of land, water, and air results in wind energy. Evaporation keeps the water cycle going, which allows hydroelectric energy to be harnessed. Sunlight captured by the biosphere results in carbon-based foods that provide energy for life, as well as the raw material for fossil fuels.

Big Question

How do people use sunlight as a source of energy?

Solar power is the use of sunlight to meet energy needs. A **solar cell** is a technology that converts sunlight to electricity. Multiple solar cells can be assembled into panels, which can heat buildings or charge batteries.

Vocabulary

solar power, n. the use of sunlight to meet energy needs

solar cell, n. a device that converts sunlight to electricity



Each of these shiny panels is a solar cell that converts sunlight into electricity.

Solar Energy's Costs and Benefits

Every hour, enough solar energy strikes Earth's surface to meet humanity's energy needs for an entire year. With that much renewable energy available, why aren't we using solar energy for everything? One challenge is the problem of solar cells. For decades, solar cells, also called photovoltaic cells, were inefficient. They could only convert about six percent of the energy in sunlight that struck them into electricity. Since then, efficiency has improved to about twenty percent. Higher-efficiency cells are made, and they are expensive. They are mostly used for satellites and other big projects.

Word to Know

How *efficient* something is refers to how quickly, easily, and well it does what it is intended to do. For example, if it takes more work and energy to make a solar cell than the energy and benefit it can provide, it is not efficient.



A large solar farm can harness a lot of sunlight and provide a lot of electricity, but what happens when the sky is cloudy or the sun goes down?

It doesn't make sense to spend a lot of money on solar cells that can only convert a little energy. But recent improvements and more demand for solar cells have brought costs down. It can still cost thousands of dollars to put enough solar cells on a home to provide electricity. For most people, that is too great an expense. And they do not want to wait years for that investment to pay off in lower electricity bills.

There is another challenge to solar power. Different amounts of sunlight strike Earth in different locations and at different times. The sun does not shine in the sky all the time. So, even if a home has solar panels or a town is using a solar farm to provide electricity, there still needs to be battery storage. Batteries capture the electricity for cloudy days and sunless nights. Batteries are costly, too.



Indirect Use of Solar Energy

Solar energy can also be converted to electricity by another method. This involves reflecting many different rays of sunlight toward a single target that becomes superheated. The target is usually the top of a tall tower in the middle of a field of computer-controlled mirrors. Liquid inside the target becomes heated to generate steam. The steam powers a turbine, which produces electricity in a generator. Hot liquid can be stored in tanks and directed to the boiler later in the day when the sun is low.



The heat that is absorbed by sunlit materials can also be used to heat water for use in homes. Some homes are equipped with rooftop water heaters. A pump carries water or some other liquid to the rooftop unit, where it gets warmed by the sun. That liquid is then directed into a tank inside the home. Water in the tank is warmed by the heat from the liquid inside pipes that came from the roof. It is available for use in the kitchen sink, shower, and other places.



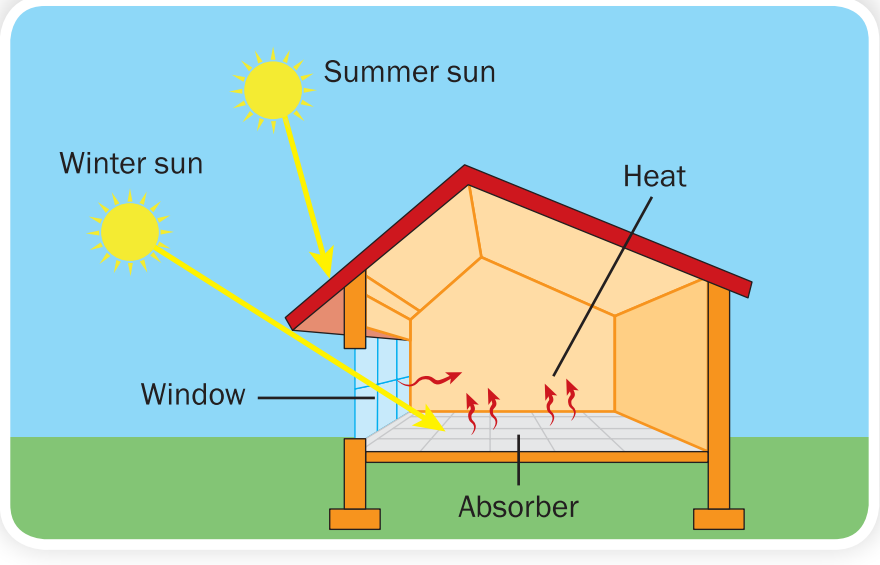
Some people also use solar ovens to cook food. The oven usually consists of a clear-roofed container that has mirrored walls. The sunlight that enters the container reflects off the walls and strikes the food inside. A solar oven usually cannot achieve the temperatures of an electric, gas, or coal-fired oven. But if enough time is given, it can cook some food very well. And the sunlight is free!

This solar thermal energy plant in California uses thousands of mirrors to reflect sunlight at a tower. The top of the tower has molten salt inside. The salt is heated and directed into a boiler, which produces steam that drives a turbine and generator.

Passive Solar Energy

Another way for people to use solar energy is to design buildings so that sunlight warms them inside. For example, a home that has a large living room that faces the sun can have large windows that allow sunlight in. The floor can be made of dark material that absorbs and retains heat and then slowly releases it throughout the day. When the sunlight is not available, insulating blinds can be drawn to prevent heat from leaving the room.

This kind of solar energy use is passive. There are no active pumps, electrical devices, or other sources of energy required. The architect or engineer studies how the sun moves through the sky at the site. They then design a structure that uses materials in the best way to let sunlight in, absorb it, and heat that structure. Such designs can also focus on ways to keep sunlight out or cool the structure down.



A solar-heated home in the Northern Hemisphere will have large windows facing south and absorbent flooring that can radiate heat over the course of the day. In summer, the roof blocks the afternoon sunlight from entering the windows and providing heat that the home does not need.